

TREE-01

Q2)

Question:

DS	Session	Tree 1	Question Information	Level 1	Challenge 62
Question description					
There are three common ways to traverse the nodes of a binary tree:					
• <i>Preorder</i>: First process the root, then the left subtree, and finally the right subtree. • <i>Inorder</i>: First process the left subtree, then the root, and finally the right subtree. • <i>Postorder</i>: First process the left subtree, then the right subtree, and finally the root.					
There is a binary tree of n nodes with distinct labels. You are given the preorder and inorder traversals of the tree, and your task is to determine its postorder traversal.					
Constraints					
• $1 \leq n \leq 10^5$					
Input					
The first input line has an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.					
After this, there are two lines describing the preorder and inorder traversals of the tree. Both lines consist of n integers.					
You can assume that the input corresponds to a binary tree.					

PROGRAM:

```
1  #include <stdio.h>
2
3  int pre[100005], in[100005], pos[100005];
4  int idx;
5
6  void post(int l, int r)
7  {
8      if (l > r)
9          return;
10
11     int root = pre[idx++];
12     int p = pos[root];
13
14     post(l, p - 1);
15     post(p + 1, r);
16     printf("%d ", root);
17 }
18
19 int main()
20 {
21     int n, i;
22
23     scanf("%d", &n);
24
25     for (i = 0; i < n; i++)
26         scanf("%d", &pre[i]);
27
28     for (i = 0; i < n; i++) {
29         scanf("%d", &in[i]);
30         pos[in[i]] = i;
31     }
32
33     post(0, n - 1);
34
35     return 0;
36 }
```

OUTPUT:

TEST CASE 1:

```
5
5 3 2 1 4
3 5 1 2 4
3 1 4 2 5
```

TEST CASE 2:

```
6
6 5 3 2 1 4
6 3 5 1 2 4
3 1 4 2 5 6
```